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Not All Sparsity Is Equal: A Controlled Evaluation of Recommender Systems under Limited User Feedback

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***Abstract*—Recommender systems rely on historical user interactions to infer preferences, but in practical settings this feedback is often limited. This study investigates how recommendation performance changes when user histories become sparse, with particular attention to whether the temporal position of retained interactions affects robustness. Ten recommender systems were evaluated on MovieLens-1M and Amazon Video Games using three controlled sparse-feedback mechanisms: random per-user, recent-history, and early-profile retention at nominal 50%, 25%, and 10% of each user's training history. Validation and test interactions were held fixed across conditions, and models were evaluated using full-sort ranking with Normalized Discounted Cumulative Gain at 10 as the primary metric.**

Performance generally declined as feedback was reduced, but the extent of degradation depended strongly on the retention mechanism and dataset. Recent-history retention preserved substantially more recommendation quality than random or early-profile retention, particularly on MovieLens, while Amazon remained more sensitive under severe sparsity because its user histories were much shorter. SASRec and GRU4Rec achieved the strongest absolute performance across most conditions, although greater architectural complexity did not consistently imply greater robustness. Overall, the findings show that sparse-feedback robustness depends not only on how much interaction data remains, but also on which interactions remain and the characteristics of the dataset.

***Index Terms*—recommender systems, data sparsity, implicit feedback, sequential recommendation, robustness evaluation, temporal retention**

I. Introduction

Recommender systems are widely used to help users navigate large collections of items by learning from their previous interactions. In many practical settings, these interactions are implicit signals such as clicks, purchases, views, or positive ratings rather than direct statements of preference [1]. Modern recommender systems use a wide range of approaches to exploit this information, from neighbourhood and latent-factor methods to neural, sequential, and graph-based models. Despite their different architectures, these methods share an important dependency: useful recommendations generally require sufficient historical information about users and items.

In practice, however, interaction histories are often sparse. Some users interact with only a small number of items, while even active users may provide only limited evidence about their current preferences. This creates a challenge for collaborative recommendation because models must infer preferences from increasingly incomplete information. Previous work has shown that recommendation performance can deteriorate as interaction data becomes sparse [2]. At the same time, methodological studies have demonstrated that conclusions about recommender performance can depend strongly on choices such as preprocessing, data splitting, baseline tuning, and evaluation procedure [3, 4]. Consequently, a model that performs strongly on a conventional full-data benchmark cannot automatically be assumed to remain strong when the available feedback is substantially reduced.

A further issue is that sparsity is not characterised only by the *number* of available interactions. User histories are temporally ordered, and interactions at different positions may contain different amounts of information about a user's current preferences. Betello et al. [5], for example, showed that sequential recommender performance is sensitive to the position of interactions removed from training sequences. This raises a broader question: when a user history becomes sparse, does it matter which portion of that history remains? Comparing models only under full interaction histories, or reducing data without considering temporal position, may therefore hide important differences in how recommender systems respond to limited feedback.

This study investigates this problem through a *controlled sparse-feedback robustness evaluation*. Ten recommender systems spanning non-personalised, classical, linear, neural, sequential, and graph-based approaches are evaluated on MovieLens-1M and the Amazon Reviews 2023 Video Games dataset. For each dataset, a fixed temporal validation and test split is created before the training histories are reduced. Three sparsification mechanisms are then applied at nominal retention levels of 50%, 25%, and 10%: *random per-user retention*, which retains interactions from across the training history; *recent-history retention*, which preserves the most recent

interactions; and *early-profile retention*, which preserves the earliest interactions. The same held-out targets and candidate catalogue are maintained across conditions so that differences can be attributed more directly to the information retained in the training histories. This controlled design also follows recent calls for stronger methodological standards and transparent offline recommender evaluation [6].

The study addresses the following research questions:

RQ1: How does recommendation performance change as the amount of available user feedback decreases?

RQ2: How does the temporal location of retained feedback affect recommendation performance under sparse user histories?

RQ3: How do different recommender approaches compare under sparse feedback, and do more complex models consistently provide greater robustness?

The overall aim is to determine how recommendation quality and model robustness change when user histories are systematically reduced. To achieve this, the study evaluates performance across progressively sparse conditions, compares the effect of different temporal retention mechanisms, and examines whether model rankings and robustness patterns generalise across two datasets with substantially different interaction-history characteristics.

The work makes three main contributions: (1) a controlled evaluation protocol that varies both the quantity and temporal location of retained user feedback while holding evaluation targets and the item catalogue fixed; (2) a cross-paradigm comparison of ten recommender systems under identical sparse-feedback conditions; and (3) empirical evidence showing that sparse-feedback behaviour depends jointly on temporal position, model choice, and dataset characteristics, rather than on interaction quantity alone.

The remainder of this paper is organised as follows. Section II reviews related work on recommender models, evaluation methodology, and sparse-feedback robustness. Section III describes the datasets, preprocessing, sparsification procedure, models, and evaluation protocol. Section IV presents the experimental results and statistical analysis. Section V discusses the findings in relation to the research questions and prior work, including limitations and future directions. Section VI concludes the paper.

II. Related Work

A. Recommender Approaches and Sequential Feedback

Recommender systems commonly learn user preferences from explicit or implicit interaction data. In implicit-feedback settings, user actions such as purchases, clicks, or positive ratings provide evidence of preference without directly observing negative feedback [1]. Classical collaborative-filtering approaches model these interactions through neighbourhood relationships or latent representations, with methods such as Bayesian Personalized Ranking (BPR) providing a widely used framework for learning from implicit preferences [7]. These approaches generally treat a user's interaction history as a set of preferences rather than explicitly modelling when each interaction occurred.

Sequence-aware recommendation extends this formulation by treating interaction order as informative [8]. Recurrent approaches such as GRU4Rec [9] model evolving behaviour through recurrent hidden states, while SASRec [10] uses self-attention to identify relevant interactions within a user's history. BERT4Rec [11] further applies bidirectional Transformer-based sequence modelling. Graph-based methods provide another approach to collaborative information, with LightGCN [12] propagating user and item representations through the interaction graph. Together with neural collaborative-filtering and autoencoder approaches, these developments provide increasingly diverse ways of using interaction history. However, stronger model complexity does not necessarily imply consistently better recommendation performance, particularly when comparisons involve differently tuned baselines or experimental protocols.

B. Rigorous and Reproducible Evaluation

The reliability of offline recommender comparisons has become an important research concern. Neural methods do not consistently outperform well-tuned simpler baselines [3], while preprocessing, tuning, and other evaluation choices can materially affect comparisons [4]. DaisyRec 2.0 formalised these evaluation factors [13], and recent guidance continues to emphasise tuned baselines, transparent procedures, and reproducibility [6].

Evaluation choices also affect the meaning of reported ranking metrics. Krichene and Rendle [14] showed that metrics calculated using sampled candidate sets can be inconsistent with their exact full-catalogue counterparts and recommended avoiding sampling where feasible. For sequential recommendation, the way interactions are divided between training and evaluation is similarly important. Gusak et al. [15] demonstrated that different temporal splitting strategies can change model rankings and noted that commonly used leave-one-out protocols may not fully represent a globally future-facing deployment scenario. These findings motivate clearly specifying both the candidate-ranking procedure and the temporal evaluation protocol when comparing recommender robustness.

C. Sparse Feedback and Temporal Robustness

Data sparsity has long been recognised as a challenge for collaborative recommendation because each user typically interacts with only a small fraction of the available catalogue. Silva et al. [2] directly evaluated collaborative-filtering algorithms under different sparsity levels and showed that changes in available interaction data can materially affect recommendation performance. Much sparse-recommendation research has therefore focused on developing models that can learn effectively from limited observations. However, interaction count alone does not capture the temporal structure present in user histories.

Recent work has begun to examine robustness to missing or perturbed sequential interactions more directly. Betello et al. [5] investigated the removal of interactions from different positions within ordered user sequences and found that perturbing interactions near the end of the sequence had a substantially larger effect on recommendation performance than removing

interactions from earlier positions. This suggests that the temporal location of missing information can be as important as the total amount removed.

Standardised reproduction has further shown that sequential-model conclusions, including SASRec versus GRU4Rec comparisons, can depend on experimental configuration and model capacity [16].

D. Summary and Positioning

Existing research therefore establishes three relevant points: recommendation performance is sensitive to data sparsity; temporal information can affect which interactions are most useful; and experimental design can materially influence model comparisons. However, these issues are often examined separately or within a restricted model family. The present study builds on this work through a controlled cross-paradigm evaluation of ten recommender systems. Rather than varying only the overall amount of training data, it systematically varies both the *quantity* and *temporal location* of retained user feedback through random per-user, recent-history, and early-profile retention. The same protocol is applied to MovieLens-1M and Amazon Video Games, whose substantially different user-history lengths provide a further basis for examining whether sparse-feedback robustness generalises across datasets. This positions the study as an evaluation of recommender robustness under controlled feedback reduction rather than as the proposal of a new recommendation algorithm.

III. Methodology

A. Experimental Design

This study uses a controlled experimental pipeline to evaluate how recommender systems respond when the amount and temporal position of available user feedback are changed. The central design principle was to vary the training history available to each model while keeping the recommendation task itself fixed. After preprocessing, each user's interactions were ordered chronologically and divided using temporal leave-one-out: the final interaction was used for testing, the second-last for validation, and all preceding interactions for training. Sparsification was then applied only to the training portion. Consequently, the validation and test targets remained identical across the full-data baseline and all sparse conditions. The post-preprocessing item catalogue was also kept fixed so that differences between conditions were not caused by changing candidate sets.

Figure 1 summarises the experimental pipeline. The pipeline proceeds from preprocessing and temporal splitting to controlled sparsity generation, baseline hyperparameter selection, model training, full-sort evaluation, and robustness analysis. Hyperparameters were selected using validation NDCG@10 on the full-data baseline and were then held fixed for all sparse conditions. This allows the experiments to measure the robustness of a common full-data-selected configuration rather than re-optimising each model separately for every sparsity level.

B. Datasets and Preprocessing

Experiments were conducted on MovieLens-1M [17] and the Amazon Reviews 2023 Video Games subset [18]. The datasets provide a useful contrast in interaction-history length: MovieLens contains comparatively long user histories, whereas the filtered Amazon dataset remains substantially sparser.

Explicit ratings were converted to positive implicit feedback by retaining interactions with ratings of at least four. For Amazon, repeated positive user-item pairs were deduplicated by retaining the most recent positive interaction; 50,578 repeated rows were removed. No retained positive duplicates were present in MovieLens. Both datasets were then subjected to iterative 5-core filtering so that every remaining user and item had at least five interactions.

Interactions were ordered first by timestamp and then by their original source-row order to deterministically resolve timestamp ties. A sequential order field was assigned after sorting and used throughout temporal splitting and sequential-model training. Table I reports the final dataset and split statistics.

The mean post-filtering history length was 95.19 interactions for MovieLens and 8.41 for Amazon. After holding out validation and test interactions, users therefore had an average of 93.19 training interactions on MovieLens but only 6.41 on Amazon. This difference is important because an equivalent nominal reduction in user history does not leave the same amount of usable evidence in the two datasets.

C. Sparse-Feedback Construction

Three sparsification mechanisms were applied independently to each user's training history at nominal retention levels of 50%, 25%, and 10%, in addition to the 100% baseline.

Random per-user retention samples interactions from across the user's training history and is intended to reduce interaction quantity without deliberately favouring a particular temporal region. A fixed sparsity seed of 2025 was used, and the retained subsets were nested so that the 10% subset was contained within the 25% subset, which was contained within the 50% subset.

Recent-history retention preserves the most recent training interactions and tests whether a shorter but temporally relevant history can retain sufficient information for predicting the user's next interaction.

Early-profile retention preserves the earliest training interactions. This represents a cold-start-like setting for known users, but it is not a true new-user cold-start experiment because the same users are present in training. It also increases the temporal distance between the retained training interactions and the held-out test target.

For a user u with n_u training interactions and nominal retention rate p, the number retained was

$$m_u(p) = \max\left(1, \lceil n_u p \rceil\right). \tag{1}$$

The minimum of one retained interaction in (1) prevents users from disappearing from the training set. As a result, nominal retention differs from aggregate realised retention, particularly for Amazon because its histories are much shorter. MovieLens realised retention was 50.270%, 25.404%, and 10.479%, while Amazon realised retention was 54.712%,

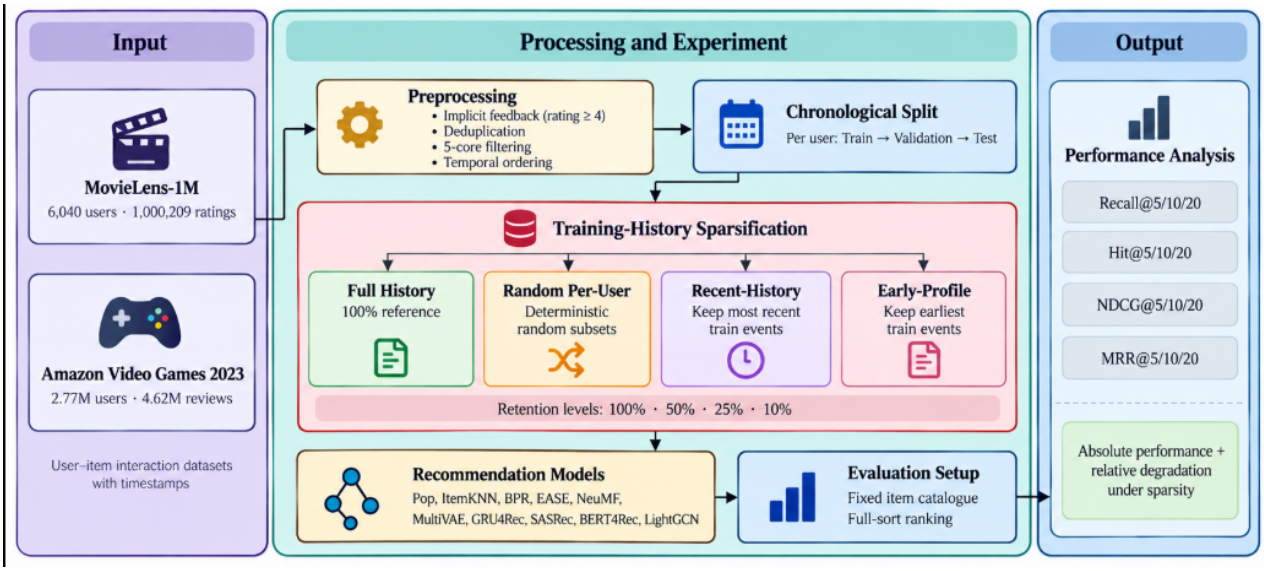


Fig. 1. Experimental pipeline. Explicit feedback is converted to positive implicit interactions, filtered and chronologically ordered before a per-user temporal leave-one-out split. Only the training histories are sparsified using random per-user, recent-history, or early-profile retention. Models are tuned on the full-data baseline, the selected configurations are frozen, and all conditions are evaluated using the same validation/test targets and fixed item catalogue.

TABLE I
FINAL DATASET AND TEMPORAL SPLIT STATISTICS.

Dataset	Users	Items	Interactions	Train	Validation	Test	Median history
MovieLens-1M	6,034	3,125	574,376	562,308	6,034	6,034	58
Amazon Video Games	63,413	19,022	533,550	406,724	63,413	63,413	6

29.767%, and 18.380%. At the nominal 10% setting, this corresponds to an average of 9.76 retained training interactions per MovieLens user but only 1.18 per Amazon user.

D. Recommender Models and Training

Ten recommenders were selected to cover a range of commonly used recommendation paradigms rather than focusing on a single model family. These were the non-personalised popularity baseline Pop; ItemKNN [19] and BPR [7] as classical approaches; the linear EASE model [20]; NeuMF [21] and MultiVAE [22] as neural collaborative-filtering approaches; GRU4Rec [9], SASRec [10], and BERT4Rec [11] as sequential recommenders; and LightGCN [12] as a graph-based recommender.

Experiments were implemented using RecBole 1.2.1 [23]. Hyperparameters were selected on the full-data baseline using validation NDCG@10, and the selected configurations were frozen across sparse conditions so that the experiment measures the robustness of a common full-data-selected configuration. The maximum sequence length was 200 interactions for MovieLens and 50 for Amazon.

Stochastic models (BPR, NeuMF, MultiVAE, GRU4Rec, SASRec, BERT4Rec, and LightGCN) were evaluated using seeds 2025, 2026, and 2027, while deterministic models (Pop, ItemKNN, and EASE) were run once per condition.

E. Evaluation Metrics

All models were evaluated using RecBole full-sort ranking rather than sampled-negative evaluation [14]. Each user's held-out test interaction was ranked against the fixed candidate catalogue. Results were recorded for Recall, HitRate, NDCG, and MRR at $K \in \{5, 10, 20\}$, with NDCG@10 used as the primary metric because it captures both successful retrieval and the rank assigned to the relevant item.

Because temporal leave-one-out produces exactly one relevant test item per user, the user-level NDCG contribution at cutoff K [24] can be written as

$$\mathrm{NDCG@}K = \begin{cases} \dfrac{1}{\log_2(r+1)}, & r \le K, \\ 0, & r > K, \end{cases} \tag{2}$$

where r is the rank of the held-out item. Under the same design, Recall@K and HitRate@K are numerically identical and are therefore not interpreted as independent evidence.

To compare robustness between models with different baseline accuracies, relative NDCG change for model m in sparse condition c was calculated as

$$\Delta_{m,c} = \frac{S_{m,c} - S_{m,\mathrm{full}}}{S_{m,\mathrm{full}}} \times 100. \tag{3}$$

Here, $S_{m,c}$ is sparse-condition NDCG@10 and $S_{m,\mathrm{full}}$ is the corresponding full-data score. Negative values from (3) indicate degradation and positive values improvement. Relative changes are considered together with absolute NDCG because

a model can appear relatively stable while still providing low recommendation quality.

F. Statistical Analysis

Statistical analysis was performed on aggregate model-level results. For stochastic models, the three final seeds were averaged before the main inferential comparisons so that repeated training runs were not incorrectly treated as independent observations. Deterministic models contributed their single result per condition.

A significance level of $\alpha = 0.05$ was used. Friedman tests were used for omnibus comparisons involving multiple related models or experimental conditions, with Kendall's W reported as an effect-size measure. Where the overall model comparison was significant, Nemenyi post-hoc comparisons were used to examine differences in average model ranks following the multiple-algorithm comparison framework described by Demšar [25]. For focused paired comparisons between sparsity mechanisms or retention levels, Wilcoxon signed-rank tests were used with Holm adjustment for multiple comparisons [26] and rank-biserial correlation as an effect-size measure. Spearman's rank correlation was used to assess agreement between model rankings and robustness rankings across datasets.

These tests support conclusions about systematic differences across the evaluated models and experimental conditions. They are not interpreted as user-level significance tests between individual recommenders, since per-user recommendation ranks were not retained for the final aggregate analysis.

IV. Experiments and Results

A. Experimental Setup

Ten recommender systems were evaluated on MovieLens-1M and Amazon Video Games under the full-data baseline and nine sparse-feedback conditions. Random per-user, recent-history, and early-profile retention were each evaluated at nominal retention levels of 50%, 25%, and 10%. NDCG@10 is the primary metric, with stochastic-model results reported as means across seeds 2025, 2026, and 2027. The complete experimental protocol is defined in Section III.

B. Full-Data Baseline Performance

Table II presents baseline NDCG@10 performance before any training interactions were removed. SASRec achieved the highest baseline score on both datasets, followed by GRU4Rec. On MovieLens, the three sequential models formed a clear leading group, with SASRec obtaining 0.1191, GRU4Rec 0.1126, and BERT4Rec 0.1067. The strongest non-sequential model, EASE, achieved 0.0452.

The differences were smaller on Amazon. SASRec again ranked first with an NDCG@10 of 0.0463, followed by GRU4Rec at 0.0449 and EASE at 0.0406. Across all 20 dataset-condition combinations considered in the study, SASRec achieved the highest mean NDCG@10 in 17 conditions, while GRU4Rec ranked first in the remaining three.

TABLE II
Full-data baseline NDCG@10 performance. Values are mean ± standard deviation for stochastic models and single-run scores for deterministic models.

Model	MovieLens-1M	Amazon Video Games
SASRec	0.1191 ± 0.0023	0.0463 ± 0.0004
GRU4Rec	0.1126 ± 0.0015	0.0449 ± 0.0010
BERT4Rec	0.1067 ± 0.0026	0.0391 ± 0.0003
EASE	0.0452	0.0406
MultiVAE	0.0392 ± 0.0006	0.0346 ± 0.0004
LightGCN	0.0381 ± 0.0007	0.0359 ± 0.0004
BPR	0.0380 ± 0.0012	0.0355 ± 0.0005
ItemKNN	0.0373	0.0241
NeuMF	0.0320 ± 0.0009	0.0243 ± 0.0003
Pop	0.0192	0.0163

TABLE III
Median relative NDCG@10 change from the full-data baseline.

Mechanism	Retention	MovieLens	Amazon
Random	50%	−40.6%	−33.5%
	25%	−56.6%	−61.5%
	10%	−74.9%	−63.6%
Recent	50%	−15.1%	−20.3%
	25%	−2.9%	−45.2%
	10%	−6.8%	−63.5%
Early	50%	−49.1%	−41.0%
	25%	−65.2%	−65.6%
	10%	−74.7%	−68.5%

C. Performance Under Sparse Feedback

Figure 2 compares NDCG@10 as the retained portion of each user's training history decreases. The three sparsity mechanisms produce visibly different performance trajectories on both datasets.

Random per-user retention produced substantial degradation as less feedback remained. On MovieLens, the median relative NDCG@10 change across the ten models decreased from −40.6% at 50% retention to −56.6% at 25% and −74.9% at 10%. Amazon showed a similar pattern, with median changes of −33.5%, −61.5%, and −63.6%.

Early-profile retention was also consistently damaging. At nominal 10% retention, the median change reached −74.7% on MovieLens and −68.5% on Amazon.

Recent-history retention produced a substantially different pattern. On MovieLens, median relative degradation was only −15.1% at 50%, −2.9% at 25%, and −6.8% at 10%. In contrast, Amazon recent-history retention degraded from −20.3% at 50% to −45.2% at 25% and −63.5% at 10%. Table III summarises these changes.

D. Behaviour Under Severe Sparsity

The nominal 10% condition provides the clearest comparison between sparsity mechanisms. On MovieLens, SASRec achieved NDCG@10 values of 0.0265 under random retention, 0.0547 under recent-history retention, and 0.0213 under early-profile retention. The benefit of recent retention was smaller on Amazon, where SASRec achieved 0.0200, 0.0278, and 0.0167

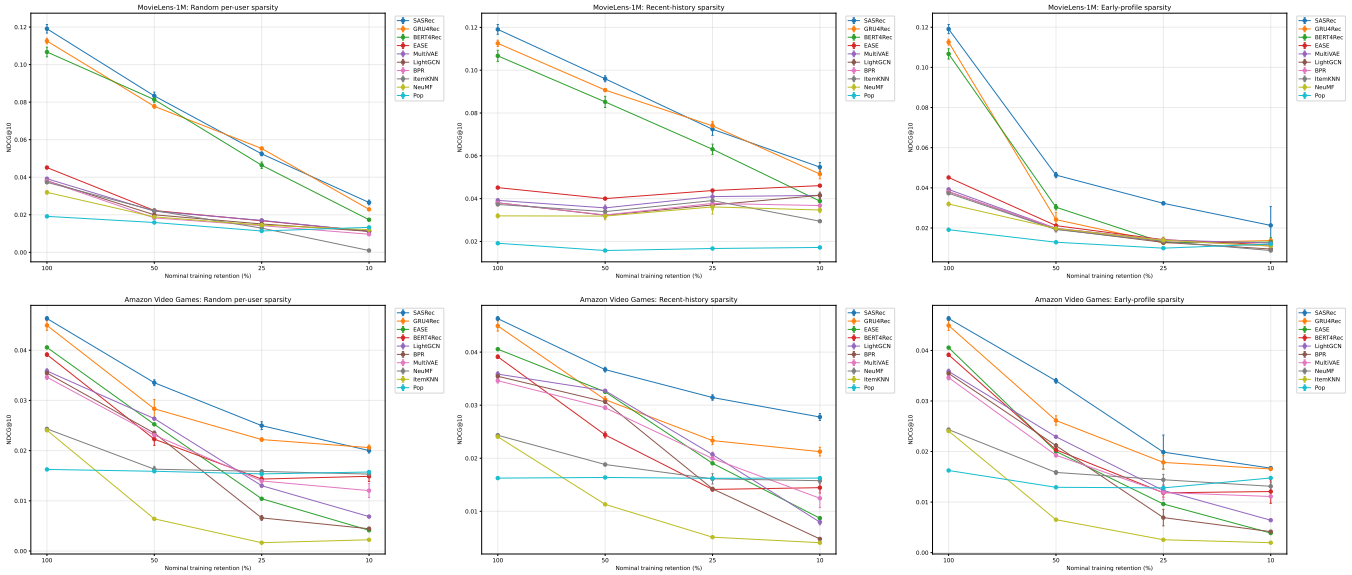


Fig. 2. NDCG@10 as training-history retention decreases under random per-user, recent-history, and early-profile sparsification. The top row shows MovieLens-1M and the bottom row Amazon Video Games. Stochastic-model points show means across three seeds; error bars indicate one standard deviation.

under the corresponding conditions. GRU4Rec was the highest-performing model under Amazon Random 10%, while SASRec remained highest under Recent and Early retention.

Relative robustness did not always correspond to the highest absolute recommendation quality. For example, Amazon Pop changed by approximately $+0.05\%$ relative to its baseline under Recent 10%, while SASRec decreased by approximately 40.0%. However, the corresponding absolute NDCG@10 values were 0.0163 for Pop and 0.0278 for SASRec. Relative degradation is therefore reported alongside absolute performance rather than used as a standalone measure of model quality.

E. Statistical Evidence

Friedman tests showed significant overall differences in NDCG@10 between the ten models on both datasets. MovieLens produced $\chi^2(9) = 63.03$, $p < 0.001$, with Kendall's $W = 0.700$, while Amazon produced $\chi^2(9) = 59.22$, $p < 0.001$, with $W = 0.658$.

Nemenyi post-hoc comparisons were then used to examine differences in average model rank. On MovieLens, SASRec had significantly better average ranks than LightGCN, NeuMF, BPR, ItemKNN, and Pop ($p < 0.05$), but its rank did not differ significantly from GRU4Rec, BERT4Rec, EASE, or MultiVAE. On Amazon, SASRec significantly outranked BERT4Rec, EASE, MultiVAE, NeuMF, BPR, ItemKNN, and Pop, while its differences from GRU4Rec and LightGCN were not significant.

The sparsity mechanism also had a significant effect at every evaluated retention level on both datasets. The six Friedman tests comparing random, recent, and early retention all produced $p < 0.001$, with Kendall's W ranging from 0.73 to 0.91. Post-hoc Wilcoxon tests showed that recent-history retention significantly outperformed both random per-user and early-profile retention at every retention level on both datasets (Holm-adjusted $p \leq 0.0078$).

When retention level itself was compared within each sparsity mechanism, five of the six dataset-mechanism combinations showed significant differences. The exception was MovieLens recent-history retention, where performance did not change systematically across 50%, 25%, and 10% retention:

$$\chi^2(2) = 1.80, \qquad p = 0.407, \qquad W = 0.09. \tag{4}$$

Table IV provides a compact summary of the main statistical results.

F. Cross-Dataset Robustness

Despite the large difference in dataset density, the full-data model rankings were highly similar. The Spearman correlation between MovieLens and Amazon baseline NDCG@10 rankings was

$$\rho = 0.939. \tag{5}$$

The agreement was much weaker when models were ranked according to their relative degradation under severe sparsity. At nominal 10% retention, the cross-dataset robustness correlations were $\rho = 0.467$ for random per-user retention, $\rho = -0.152$ for recent-history retention, and $\rho = 0.309$ for early-profile retention. None of these severe-sparsity robustness correlations was statistically significant after correction for multiple comparisons.

Supporting metrics produced similar model orderings. Across all conditions, Spearman correlations between NDCG@10 and Recall@10 rankings ranged from 0.927 to 0.988 on MovieLens and from 0.952 to 1.000 on Amazon. Agreement between NDCG@10 and MRR@10 ranged from 0.867 to 1.000 and from 0.927 to 1.000 respectively. The main model-ranking patterns were therefore not specific to NDCG@10 alone.

V. Discussion

The results show that sparse-feedback performance cannot be explained by interaction quantity alone. Although recommendation quality generally decreased as less training feedback

TABLE IV
SUMMARY OF THE MAIN STATISTICAL RESULTS.

Analysis	MovieLens-1M	Amazon Video Games
Overall model effect	$\chi^2(9) = 63.03,\ p < .001,\ W = .700$	$\chi^2(9) = 59.22,\ p < .001,\ W = .658$
Mechanism effect, 50%	$p < .001,\ W = .73$	$p < .001,\ W = .84$
Mechanism effect, 25%	$p < .001,\ W = .91$	$p < .001,\ W = .73$
Mechanism effect, 10%	$p < .001,\ W = .76$	$p < .001,\ W = .91$
Retention effect, Recent	$p = .407,\ W = .09$	$p < .001,\ W = .84$

was retained, the extent of this degradation depended strongly on which interactions remained, which model was used, and the characteristics of the dataset. This distinction is important because two conditions with the same nominal retention level could produce substantially different outcomes.

A. Effect of Decreasing User Feedback

For **RQ1**, the experiments show that reducing available user feedback generally decreases recommendation performance, but the relationship is not uniform. Random per-user and early-profile retention produced severe degradation at nominal 10% on both datasets.

Recent-history retention behaved differently. On MovieLens, performance did not change systematically across retention levels ($p = 0.407$), and several non-sequential models matched or slightly exceeded their full-history NDCG@10 when only recent interactions were retained. This does not show that removing data inherently improves recommendation; rather, some older interactions may be less informative for predicting the next held-out interaction.

Amazon did not show the same tolerance to severe history reduction. At nominal 10%, approximately 9.76 interactions per user remained on MovieLens but only 1.18 on Amazon. Recency therefore helps substantially on MovieLens but cannot compensate when very little user-specific evidence remains.

These findings are consistent with previous work showing that data sparsity can substantially affect recommender performance [2], while extending the comparison beyond overall sparsity level by explicitly controlling the temporal location of retained feedback.

B. Importance of Temporal Position

For **RQ2**, the results provide strong evidence that the temporal position of retained feedback matters. Recent-history retention significantly outperformed both random per-user and early-profile retention at every evaluated retention level on both datasets. The distinction was particularly strong on MovieLens, where the median relative degradation at nominal 10% was only 6.8% under recent retention, compared with approximately 75% under both random and early retention.

This finding is broadly consistent with Betello et al. [5], who showed that sequential recommender performance is sensitive to where interactions are removed from temporally ordered histories. Their experiments found particularly strong effects when interactions close to the end of a sequence were perturbed. Although the experimental protocols differ, both studies indicate that interactions should not be treated as equally informative simply because they belong to the same user history.

The poor performance of early-profile retention should also be interpreted carefully. It is a cold-start-like condition because only a small early portion of a known user's history is available, but it is not a true new-user cold-start experiment. In addition, the retained interactions are temporally distant from the test target. Its degradation may therefore reflect both reduced feedback quantity and stale preference information. The present experiments demonstrate that early interactions are generally less effective for the chosen next-interaction task, but they do not isolate the causal contribution of preference drift.

Sparsification also affected item exposure. As interactions were removed, some held-out test items became globally absent from the retained training data, particularly on Amazon. At early-profile 10%, this occurred for approximately 13.2% of Amazon test targets compared with 2.3% on MovieLens. The measured degradation therefore reflects both reduced user-profile information and, in some conditions, reduced opportunity for models to learn about particular items.

C. Model Performance and Robustness

For **RQ3**, SASRec and GRU4Rec provided the strongest absolute recommendation performance across the experiment. SASRec achieved the highest mean NDCG@10 in 17 of the 20 dataset-condition combinations, while GRU4Rec led in the remaining three. The significant Friedman tests further confirm that model choice had a systematic effect on performance. However, the post-hoc analysis does not support treating SASRec as statistically superior to every alternative; in particular, its aggregate rank was not significantly different from GRU4Rec on either dataset.

More importantly, the results do not support the broader claim that greater architectural complexity automatically produces greater robustness. EASE, a comparatively simple linear model, was highly competitive at the Amazon baseline and retained substantial performance under MovieLens recent-history sparsity. Conversely, more complex approaches such as NeuMF and LightGCN did not consistently outperform simpler alternatives across the sparse conditions. This is consistent with Ferrari Dacrema et al. [3], who cautioned that reported gains from increasingly complex neural recommenders can disappear when comparisons are made against appropriately configured baselines.

This result also reinforces the importance of controlled evaluation. Prior work has shown that preprocessing, splitting, parameter tuning, and evaluation choices can materially change

apparent recommender performance [4, 13], while more recent methodological guidance has emphasised the importance of tuned baselines and reproducible experimental procedures [6]. In the present study, baseline tuning was therefore performed before sparse evaluation and the selected configurations were frozen across conditions. This makes the comparison one of robustness under a consistent configuration rather than a comparison in which each sparse condition receives additional optimisation.

Absolute performance must also be separated from *relative robustness*. Pop provides the clearest example. Under Amazon Recent 10%, its NDCG@10 changed almost negligibly relative to baseline, while SASRec lost approximately 40%. However, SASRec still achieved substantially higher absolute NDCG@10. A model with little personalisation can have less performance to lose as user information disappears, so low relative degradation alone should not be interpreted as superior recommendation quality.

D. Cross-Dataset Generalisation

A further finding is that full-data model performance generalised across datasets much more strongly than sparse-feedback robustness. MovieLens and Amazon baseline model rankings had a Spearman correlation of $\rho = 0.939$, indicating that models that performed well with full histories tended to do so on both datasets. In contrast, at nominal 10% retention the cross-dataset correlations of relative robustness were only 0.467 for random retention, -0.152 for recent retention, and 0.309 for early retention, none of which was statistically significant after multiple-comparison correction.

This suggests that baseline accuracy and sparse-feedback robustness should be treated as related but distinct properties. A recommender that ranks highly on one dataset under full interaction histories cannot be assumed to degrade in the same way on another dataset when feedback becomes limited. The large difference in user-history length between MovieLens and Amazon provides one plausible explanation, although additional datasets would be required to determine how broadly this relationship generalises.

E. Limitations and Future Work

Several limitations constrain the conclusions of this study. First, iterative 5-core filtering means that the experiments evaluate controlled sparsification of users and items with established interaction histories rather than the full natural cold-start population. Second, early-profile retention combines reduced profile length with temporal staleness and should therefore not be interpreted as a pure cold-start intervention. Third, random per-user sparsity was generated from one fixed sparsity seed; although this enables controlled nested comparisons, additional independent sparsity realisations would provide stronger evidence about sensitivity to the particular interactions removed. Fourth, the minimum-one-interaction rule means that nominal retention does not correspond to the same realised data fraction across datasets, particularly on Amazon. Finally, inferential comparisons are based on aggregate model-condition results rather than per-user recommendation outcomes, limiting model-to-model significance claims at the individual-user level.

Future work could repeat random sparsification across multiple independently generated masks, record per-user held-out ranks to support paired user-level significance testing, and extend the protocol to additional domains and true cold-start settings. Retuning models separately under sparse conditions could also distinguish robustness of a fixed full-data configuration from the best achievable performance when sparsity is known in advance.

Overall, the three research questions can be answered directly. **RQ1:** recommendation performance generally decreases as user feedback becomes sparse, but the rate of degradation depends strongly on the experimental condition. **RQ2:** temporal position matters; recent interactions consistently preserve more recommendation utility than equally sized random or early-profile subsets, with the strongest effect observed on MovieLens. **RQ3:** SASRec and GRU4Rec are the strongest absolute performers among the evaluated models, but increased architectural complexity does not guarantee greater robustness. Taken together, the results show that sparse-feedback robustness is jointly shaped by feedback quantity, temporal position, model choice, and dataset characteristics.

VI. Conclusion

This study examined how recommender systems respond when the amount and temporal position of available user feedback are systematically reduced. Using a controlled sparse-feedback evaluation across MovieLens-1M and Amazon Video Games, ten recommender systems were compared under full-data conditions and under random per-user, recent-history, and early-profile retention. The results show that recommendation quality generally declines as feedback becomes sparse, but the extent of this degradation depends strongly on which interactions remain. Recent interactions usually preserve substantially more recommendation utility than equally sized random or early-profile subsets, with MovieLens showing much greater tolerance to recent-history reduction than Amazon.

Across the evaluated models, SASRec and GRU4Rec achieved the strongest absolute performance in most conditions, but greater architectural complexity did not consistently result in greater robustness. Model behaviour under sparsity also differed noticeably between the two datasets, showing that strong full-data performance does not necessarily imply similar robustness when user feedback becomes limited. Overall, the study contributes a controlled evaluation of sparse-feedback robustness and demonstrates that recommender systems should be assessed not only according to how much interaction data remains, but also according to which parts of the user history remain available.

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